Assignment - 14

1. Implement a Hash Table using Linear Probing.

Requirements:

1. Accept a fixed-size hash table from the user.
2. Insert multiple integer keys into the table using **linear probing** for collision resolution.
3. Display the final state of the hash table after all insertions.
4. Provide a search function to check whether a key exists in the table, along with the number of probes made during the search.
5. Display appropriate messages if the table is full or the key is not found.

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2. Implement a Hash Table using Chaining with Linked Lists.

Requirements:

1. Accept a hash table size from the user.
2. Use a linked list to handle collisions at each index.
3. Insert a series of keys (integers or strings) into the hash table.
4. Display the entire hash table with the contents of each chain.
5. Include a search option that returns the position (bucket index and node position) of the key if found.